Electrolux Energy Management System

UML Diagrams

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Use Case Diagram

The system has four actors and 23 use cases distributed across three role-based portals.
Full diagram: https://drive.google.com/drive/folders/1OSXGavwJ1W7aPo7mCAhk9eMlsT54XNGg?

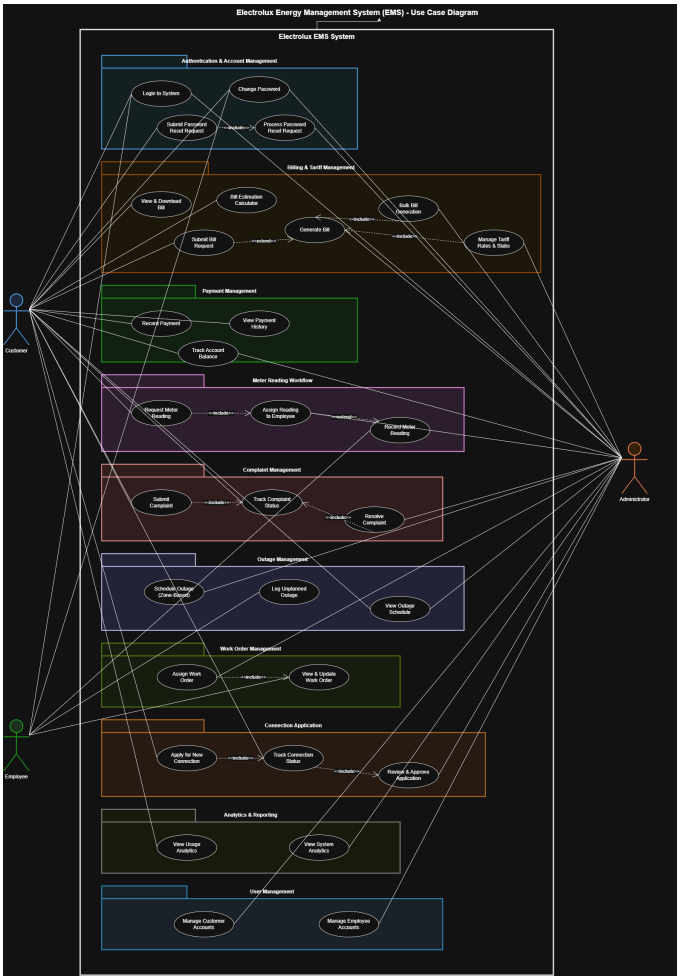

Figure 1: Electrolux EMS — Use Case Diagram (23 use cases, 4 actors)

Actor Summary

Actor	Type	Primary Use Cases
Administrator	Primary	Billing management, tariff configuration, analytics & reporting, staff management, outage scheduling, connection approvals
Customer	Primary	Bill viewing & download, payment tracking, bill calculator, usage analytics, complaint submission, outage schedule, meter reading request
Employee	Primary	Meter reading entry, work order management, bill generation for assigned customers
Public User	Primary	New connection application & status tracking (no login required)

Class Diagram

The following diagram shows the nine core domain entities, their key attributes, and their relationships.

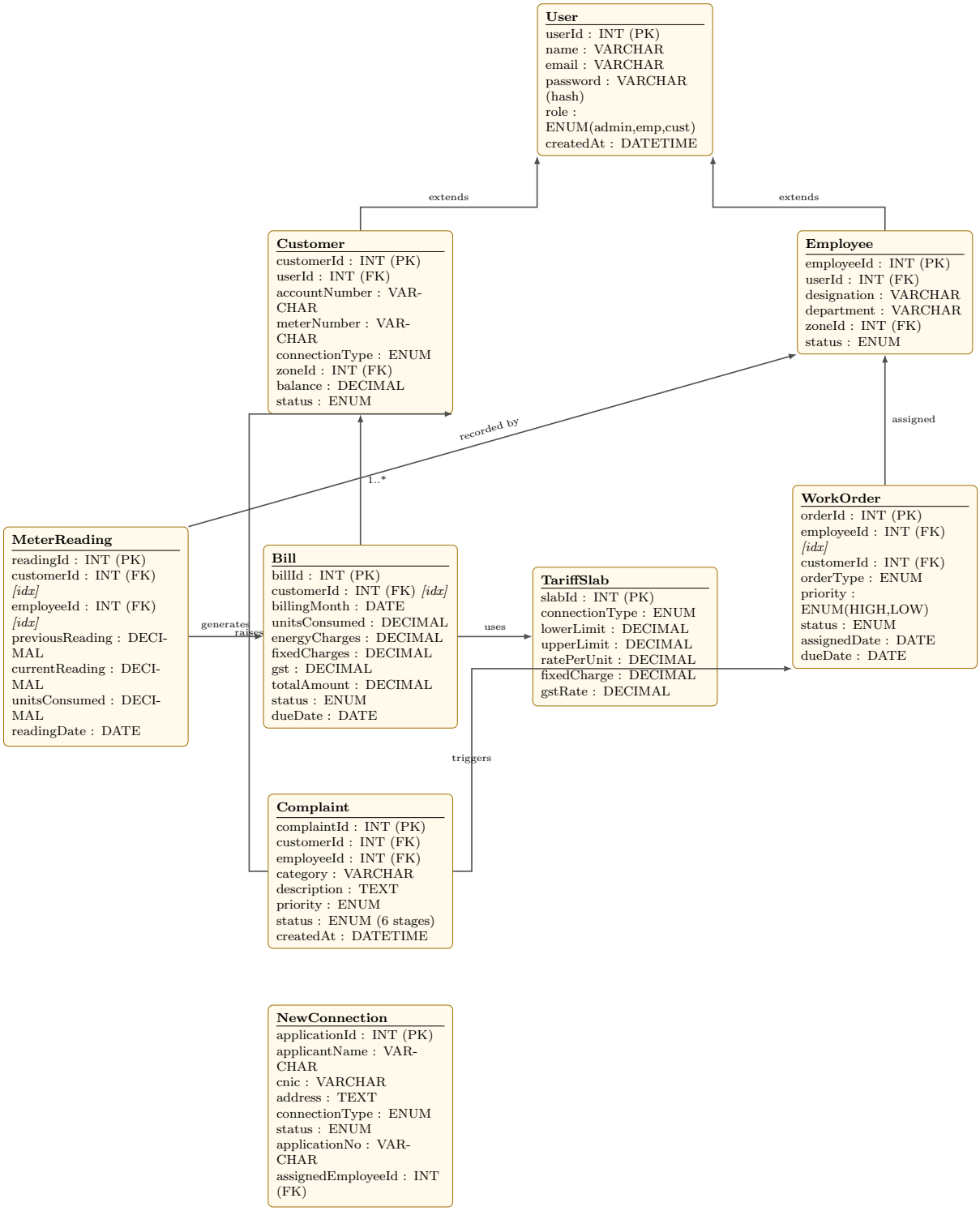

Figure 2: Class diagram — 9 core domain entities with indexed foreign keys

Sequence Diagrams

Bulk Bill Generation Sequence

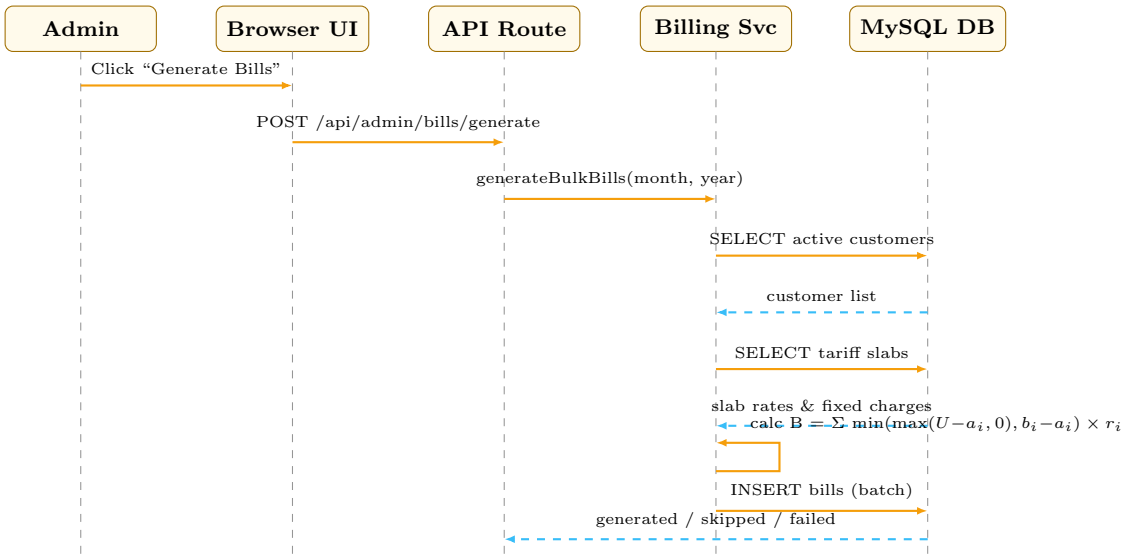

Figure 3: Sequence diagram — bulk bill generation

Login & RBAC Enforcement Sequence

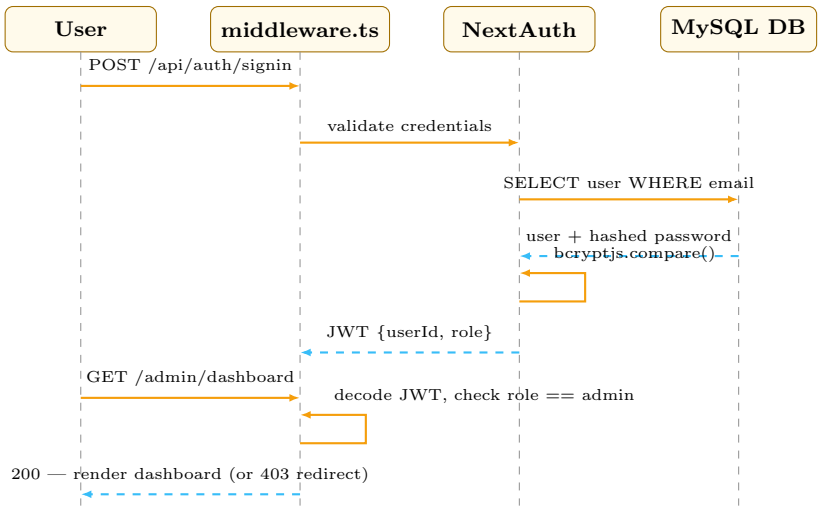

Figure 4: Sequence diagram — login authentication and RBAC enforcement via middleware.ts

Activity Diagram — New Connection Workflow

The new connection workflow spans five stages across four actors: Public User → System → Administrator → Employee → System. No login is required to apply or track an application.

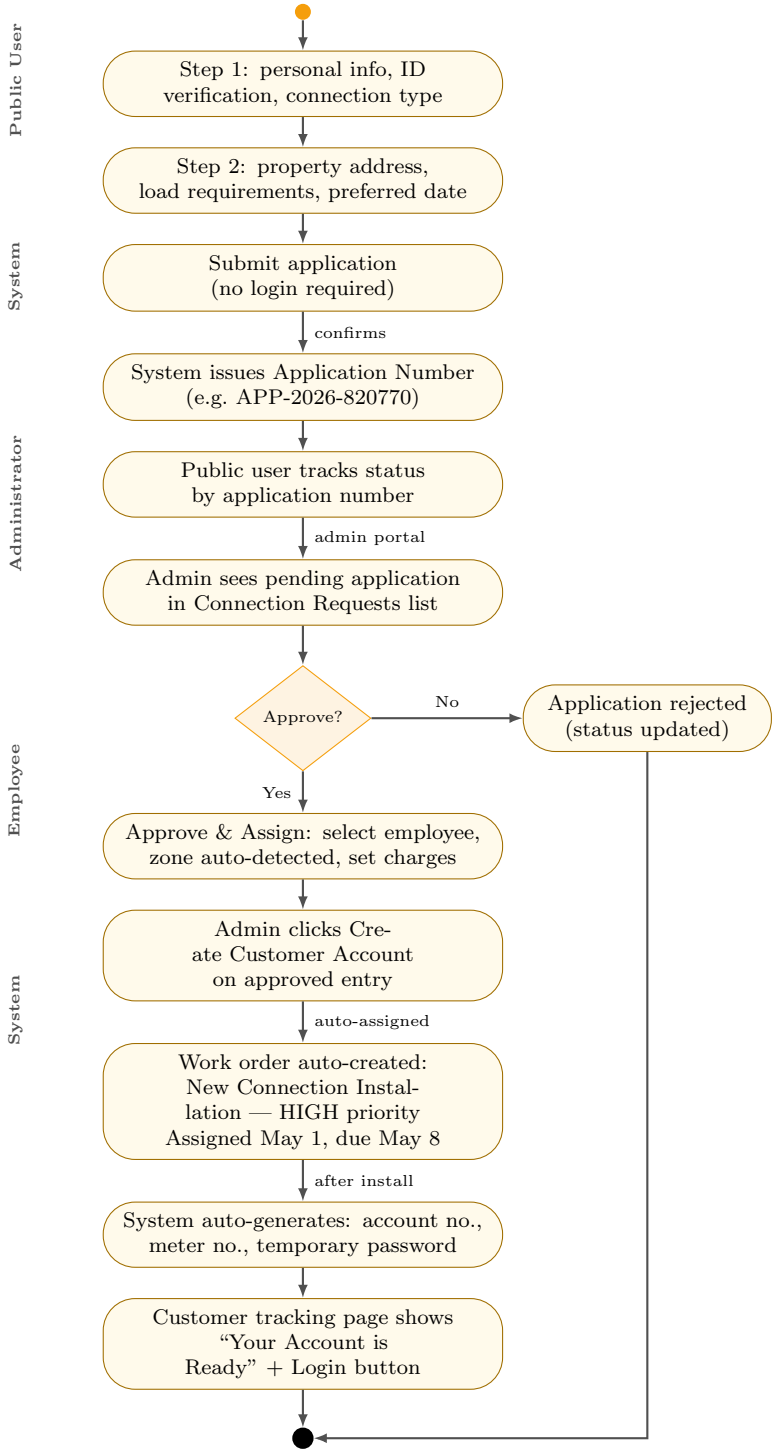

Figure 5: Activity diagram — new connection workflow (5 stages, 4 actors)

Workflow Summary Table

Stage	Actor	Action	Output
1	Public User	Fill & submit 2-step form	Application submitted (no login)
2	System	Issue application number	APP-2026-820770 issued
3	Administrator	Review, Approve & Assign employee	Work order created
4	Employee	Complete installation (HIGH priority)	New connection installed
5	System	Auto-generate customer account	ELX-2026-707493-B09B + credentials

Component Diagram

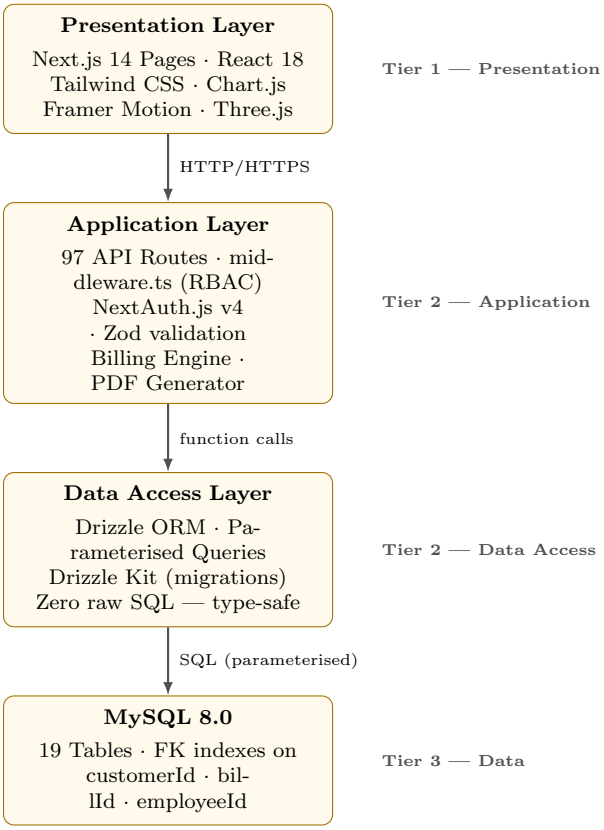

Figure 6: Component diagram — three-tier architecture

State Diagram — Complaint Lifecycle

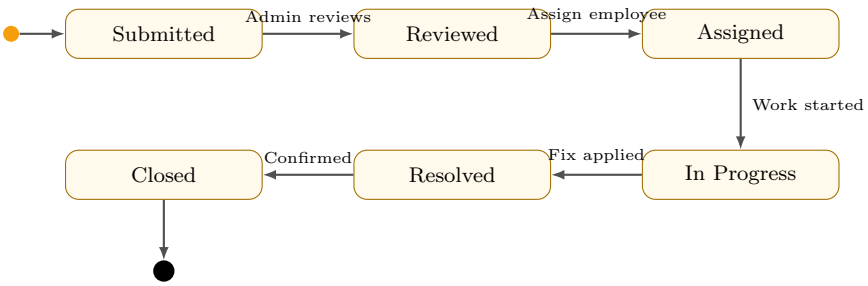

Figure 7: State diagram — complaint lifecycle (6 stages)